VIDHESH SHUKLA

Staff Geologist II · Environmental & Subsurface Consulting · Data-Driven Site Characterization

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PROFESSIONAL SUMMARY

Staff Geologist II with 3+ years of experience leading multi-state subsurface investigations and remediation programs. Expertise in contaminant plume delineation, groundwater flow analysis, and data-driven site characterization. Combines environmental consulting execution with Python/R/MATLAB-based automation and modeling. ArcGIS Pro Professional Certified, HAZWOPER 40-hr, PG license in progress

CORE COMPETENCIES

Regulatory: CERCLA/RCRA, Phase I/II ESA (ASTM E1527-21), VCP, Brownfields, NEPA, UST programs, multi-agency coordination

Site Investigation: PFAS/PFOS characterization, vapor intrusion (VI) assessment, RBCA risk-based closure, 3D plume & geological modeling, borehole logging,

Data Analysis: Python, R, MATLAB, Pandas, NumPy; geospatial integration of analytical datasets for plume delineation & stratigraphic interpretation; machine learning

Software: ArcGIS Pro (Professional Cert), QGIS, Strater, strong spatial programming base applicable to Petrel/Kingdom

Field Operations: Multi-state campaigns (AL, GA, TN, LA, FL, MS, NC, KS); drilling oversight; health & safety (HAZWOPER, OSHA 10-hr)

PROFESSIONAL EXPERIENCE

Staff Geologist II · ***United Consulting · Birmingham, AL*** ***Mar 2024 – Present***

- Led subsurface investigations across 7 states managing ~$1.5M in project value; achieved 100% regulatory acceptance and zero compliance violations across all project phases.
- Designed PFAS/PFOS sampling strategies, geotechnical investigations, asbestos surveys, and radon testing at 15+ sites; expanded firm service offerings into high-growth technical sectors.
- Developed potentiometric surface maps and integrated analytical data to assess groundwater flow direction and contaminant migration, directly informing remediation strategies and site closure decisions.
- Executed complex 3D site models utilizing 200+ soil borings and 75+ monitoring wells across diverse geological provinces (Coastal Plain, Piedmont, Appalachian).
- Authored technical reports, work plans, and remedial investigation documents with minimal agency revisions — zero requests for additional information (RAI) on managed submittals.

Project Manager / Geological Specialist · ***Kansas Dept. of Health & Environment (KDHE) · Topeka, KS*** ***Mar 2023 – Mar 2024***

- Evaluated 52 subsurface projects with ~$18M combined liability; optimized investigation designs for maximum data ROI and 100% statutory compliance.
- Managed QA/QC for 500+ annual analytical datasets; identified data gaps that prevented $2M+ in unnecessary operational expenditures.
- Developed and reviewed RBCA risk-based closure proposals; negotiated remedial strategies with responsible parties and legal counsel for complex multi-party sites.
- Authored scopes of work, RFPs, and technical specifications for field investigation contractors across VCP and Brownfields programs.
- Led community engagement for high-profile sites with 200+ affected residents, translating complex technical findings into accessible public communications.

Graduate Research Assistant · ***Kansas State University · Manhattan, KS*** ***Jan 2021 – Dec 2022***

- Developed the first high-resolution glacial chronology for Montana's Flathead Lobe using OSL dating; results incorporated into active USGS regional geological mapping initiatives.
- MATLAB and R applied for advanced age modeling and paleoenvironmental reconstruction spanning 50,000+ years.
- Presented at GSA Connects Conference, Kansas State University and New World Luminescence Dating Workshop.

Planetary Glaciology Intern · ISRO ***Physical Research Laboratory · India*** ***May – Jun 2019***

- Conducted geomorphometric analysis of Martian glacial features using HiRISE high-resolution satellite imagery and ArcGIS — remote sensing workflows directly applicable to large-scale environmental site mapping.

Glaciology Intern · Wadia ***Institute of Himalayan Geology · India*** ***Jun 2018 – Mar 2019***

- Quantified Himalayan glacier retreat using 30-year hydrological datasets, documenting 18% area reduction at 4,500 m elevation; developed proficiency in multi-temporal remote sensing and spatial analysis.

EDUCATION

M.S. Geology | GPA: 3.8 / 4.0 • Kansas State University, 2023 - *Thesis: Extent and glacial history of the Cordilleran Ice Sheet in northwest Montana: Using OSL to date glacial sediments from the southeastern Flathead Lobe - results utilized by USGS*

Coursework: Economic Geology, Isotope Geochemistry, Hydrogeology, Field Geophysics, Remote Sensing, MATLAB

M.S. Applied Geology • Barkatullah University, India, 2018 — *Focus: Glacial sediment dynamics*

B.S. Triple Major — Geology, Physics, Mathematics • Barkatullah University, India, 2015

CERTIFICATIONS

PG License — In Progress • ArcGIS Pro Professional Certified • Copernicus/ESA Earth Observation • OSHA HAZWOPER 40-hr • OSHA 10-hr • First Aid/CPR